Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Week 4

Lecture 7

7.1 Binary notation with multiple qubits
7.2 Quantum Registers
7.3 Hadamard gates and equal superposition
7.4 Quantum “Parallelism”
7.5 Multiqubit measurement
7.6 Toffoli gate

Lecture 8

8.1 Classical digital logic and universality
8.2 Reversible logic
8.3 Arithmetic operations on a quantum computer
8.4 Universality in quantum computing

Practice class 4

Multi-qubit states and operations

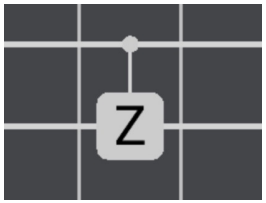
Lecture 6 recap: two-qubit logic and entanglement

CNOT

Rule: The target is flipped iff the control qubit is “1”.

$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$
$$\rightarrow a\,|00\rangle + b\,|01\rangle + d\,|10\rangle + c\,|11\rangle$$

Controlled-Z (CZ)



Rule: the **phase** of the target flipped iff the control qubit is “1”.

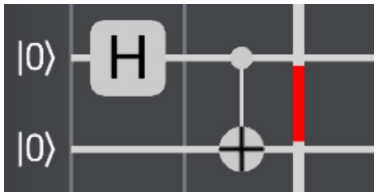
$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$
$$\rightarrow a\,|00\rangle + b\,|01\rangle + c\,|10\rangle - d\,|11\rangle$$

SWAP

Rule: the qubits **states** are swapped.

$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$
$$\rightarrow a\,|00\rangle + c\,|01\rangle + b\,|10\rangle + d\,|11\rangle$$

Entanglement:



e.g. Bell-state can’t be written as a product

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq (a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle)$$

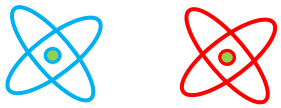
QUI: computes entanglement entropy (EE)

7.1 Binary notation with multiple qubits

MULT20015

Lecture 7

Recall from Lecture 1: Multi qubits – state counting

		Binary combinations
1 qubit:	$\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0\rangle, \lvert 1\rangle$ binary representation of decimals 0 to 1
2 qubits:	 $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 00\rangle, \lvert 01\rangle, \lvert 10\rangle, \lvert 11\rangle$ binary representation of decimals 0 to 3
3 qubits:	$\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 000\rangle, \lvert 001\rangle, \lvert 010\rangle, \lvert 011\rangle$ $\lvert 100\rangle, \lvert 101\rangle, \lvert 110\rangle, \lvert 111\rangle$ binary representation of decimals 0 to 7
4 qubits:	$\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0000\rangle, \lvert 0001\rangle, \lvert 0010\rangle, \lvert 0011\rangle$ $\lvert 0100\rangle, \lvert 0101\rangle, \lvert 0110\rangle, \lvert 0111\rangle$ $\lvert 1000\rangle, \lvert 1001\rangle, \lvert 1010\rangle, \lvert 1011\rangle$ $\lvert 1100\rangle, \lvert 1101\rangle, \lvert 1110\rangle, \lvert 1111\rangle$ binary representation of decimals 0 to 15

Binary Notation with n qubits

Binary	Decimal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Quantum computers, like classical computers, store numbers in **binary**.

Left: All the states possible for three bit/qubits.

If there are n qubits, then the number of possible states is:

$$N = 2^n$$

Specifically the integers,

$$0 \,..\, 2^n - 1$$

Binary and Decimal Notation

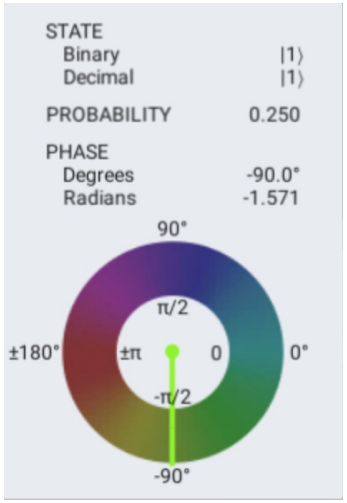
In general we use two representations in the QUI ($N = 2^n$):

Binary: $$|\psi\rangle = a_{0...00}\,|0...00\rangle + a_{0...01}\,|0...01\rangle + a_{0...10}\,|0...10\rangle + ... + a_{1...1}\,|1...1\rangle$$

Decimal: $$|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle + a_2\,|2\rangle + ... + a_{N-1}\,|N-1\rangle$$

$$|\psi\rangle = \sum_{\phi} a_\phi\,|\phi\rangle$$

$$a_\phi = |a_\phi| e^{i\theta_\phi}$$



Memory needed on a classical computer

If we use 128 bits (16 B) per complex number, then we need:

Qubits	Dimension (2^n)	Memory required
1	2	256 bits = 32 B
2	4	64 B
6	64	1 kB
10	1,024	16 kB
20	1,048,576	16 MB
30	1,073,741,824	16 GB
40	1,099,511,627,776	16 TB
50	1,125,899,906,842,624	16 PB
60	1,152,921,504,606,846,976	16 EB

Memory required to store the quantum state of different numbers of qubits **grows exponentially**.

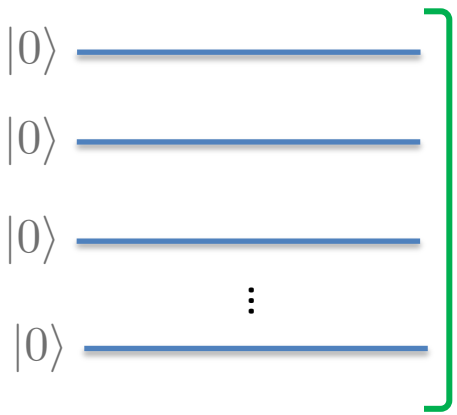
7.2 Quantum Registers

MULT20015

Lecture 7

Quantum Registers

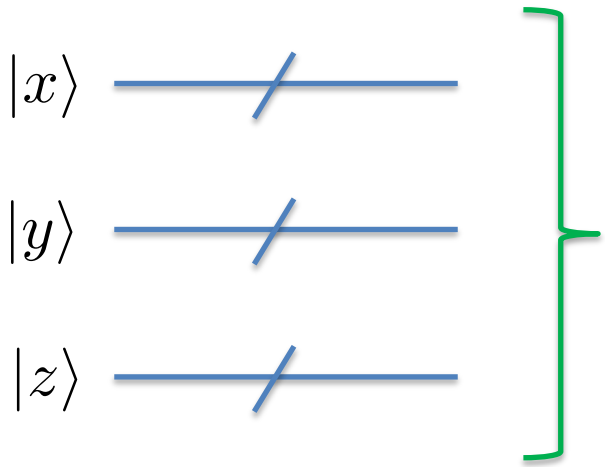
We will often need to consider several different qubits grouped together. A group of several qubits is known as a **register**.



Several qubits

Quantum register

Multiple registers



$$|\psi\rangle = |x\rangle \otimes |y\rangle \otimes |z\rangle$$

or

$$|\psi\rangle = |x\rangle\,|y\rangle\,|z\rangle$$

And of course you can have *superpositions* over these states as well. An example:

$$|\psi\rangle = \frac{1}{2}\,|1\rangle\,|5\rangle + \frac{1}{2}\,|1\rangle\,|6\rangle + \frac{1}{\sqrt{2}}\,|3\rangle\,|8\rangle$$

As before, if

$$|\psi\rangle = \sum_{ij} a_{ij}\,|i\rangle\,|j\rangle$$

then

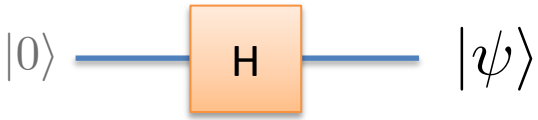
$$\sum_{i,j} |a_{ij}|^2 = 1$$

7.3 Hadamard gates and equal superposition

MULT20015

Lecture 7

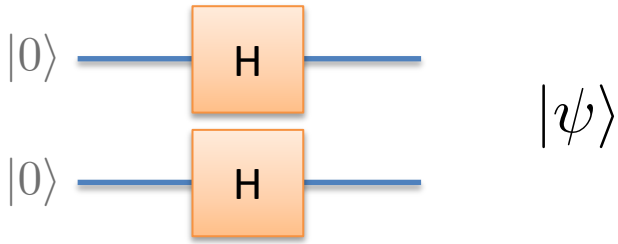
One Hadamard Gate



As we've seen before, this produces an equal superposition:

$$\begin{aligned} |\psi\rangle &= H\,|0\rangle \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \end{aligned}$$

Two Hadamard Gates

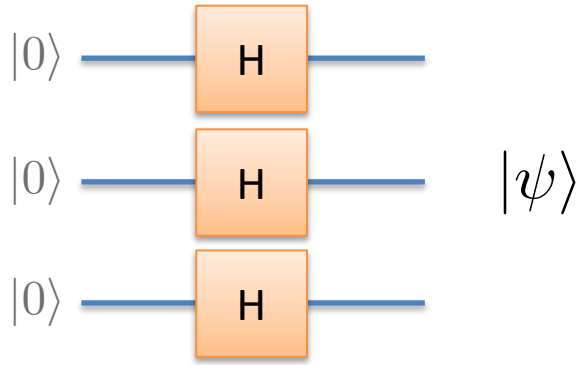


This produces:

$$
\begin{aligned}
|\psi\rangle &= H \otimes H\,|00\rangle \\
&= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \\
&= \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} \\
&= \frac{|0\rangle + |1\rangle + |2\rangle + |3\rangle}{2}
\end{aligned}
$$

We have used digital notation here

Three Hadamard Gates

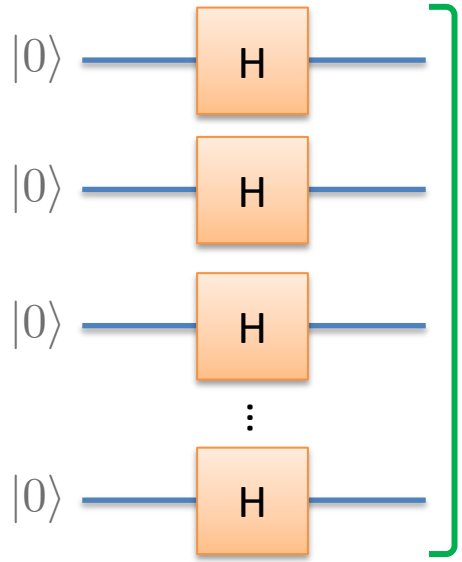


This produces:

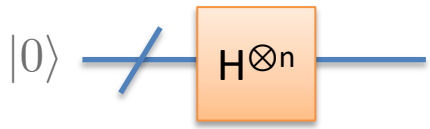
$$
\begin{aligned}
|\psi\rangle &= H \otimes H \otimes H \;\; |0\rangle\,|0\rangle\,|0\rangle \\
&= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \\
&= \frac{|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle}{2\sqrt{2}} \\
&= \frac{|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle}{2\sqrt{2}}
\end{aligned}
$$

Hadamard gates and decimal notation

n qubits



shorthand notation



$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \ldots \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|\psi\rangle = \left[\frac{1}{\sqrt{2}}\right]^n (|00\ldots0\rangle + \ldots + |11\ldots1\rangle)$$

i.e. even superposition over binary rep of integers: i = 0 to $2^n - 1$

$$|\psi\rangle = \frac{1}{2^{n/2}} \sum_{i=0}^{2^n-1} |i\rangle$$

Equal superposition of states

7.4 Quantum “Parallelism”

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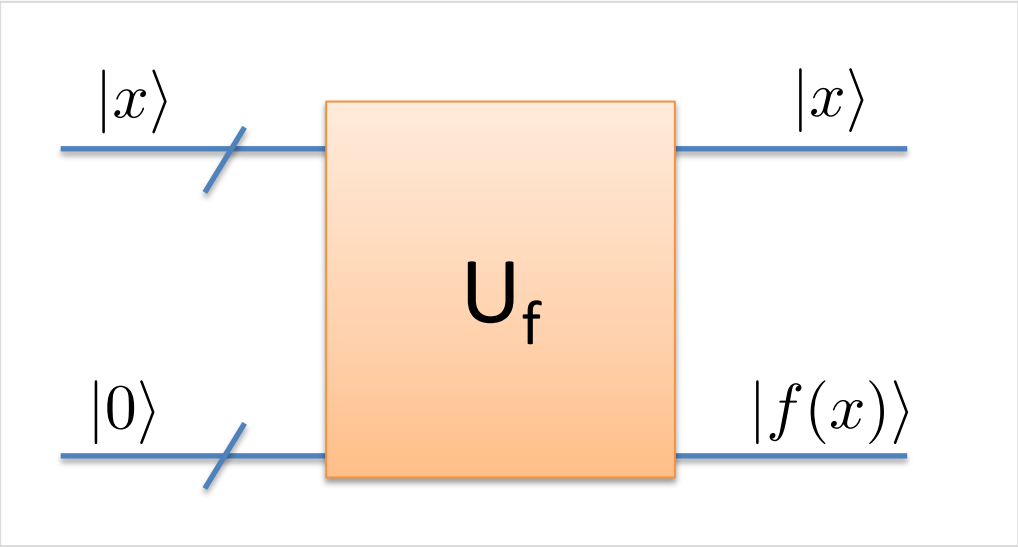
Lecture 7

Calculating some function, f

Imagine that we had written some routine/circuit to calculate some function, f.

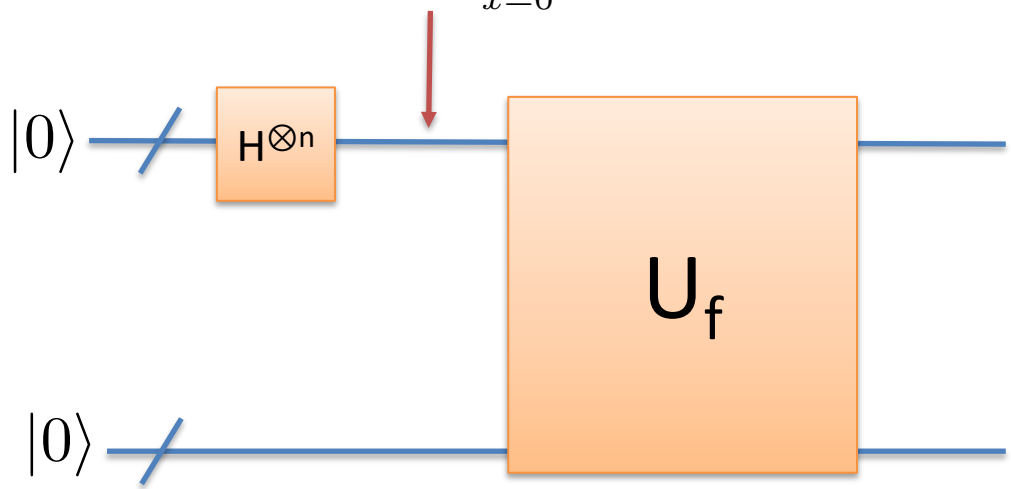


On a classical computer, we can only evaluate the function for a single input, x, at a time. What would happen if we input a superposition in a quantum computer?

Quantum Parallelism

An equal superposition of inputs

$$|\psi\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle$$



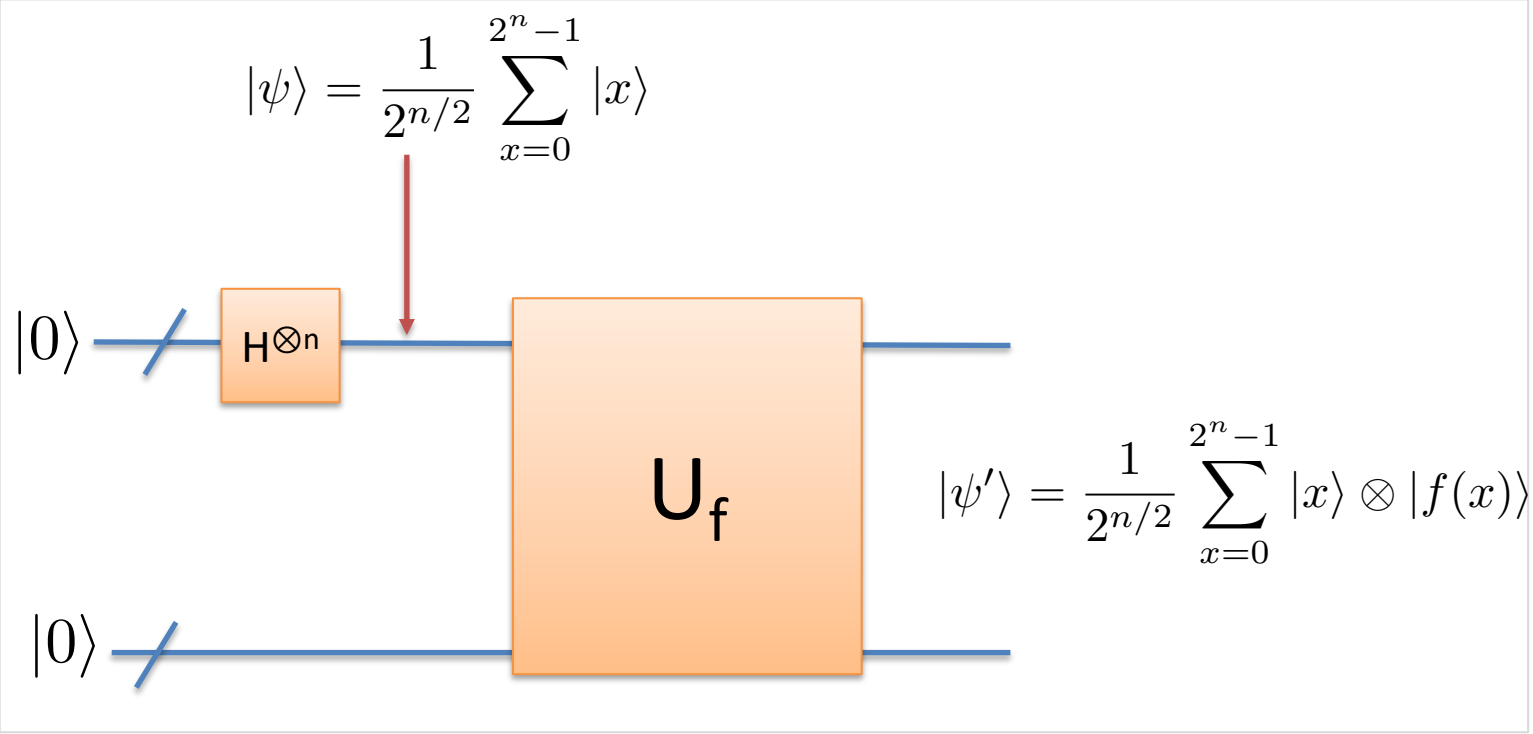
$$|\psi'\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle$$

The function evaluates all the inputs in superposition with *just one* application of U_f.

Quantum Parallelism Example

Consider evaluating the function: $f(x) = x^2$

x	f(x)
0	0
1	1
2	4
3	9
4	16
5	25
6	36
7	49



The output also is a superposition, evaluated for each input:

$$|\psi'\rangle = \frac{|0\rangle|0\rangle + |1\rangle|1\rangle + |2\rangle|4\rangle + |3\rangle|9\rangle + |4\rangle|16\rangle + |5\rangle|25\rangle + |6\rangle|36\rangle + |7\rangle|49\rangle}{2\sqrt{2}}$$

"Quantum parallelism" – evaluated on entire superposition in one application of the function.

7.5 Multiqubit Measurement

MULT20015

Lecture 7

Recap: Two-qubit measurement

Measurement on a two-qubit state:

1) Apply collapse principle to the measured state
2) Renormalize the state probabilities

For example, consider the general two-qubit state:

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$

If the first qubit were measured to be "0", keep only those states and renormalize to get the new state:

$$|\psi'\rangle = \frac{a\,|00\rangle + b\,|01\rangle}{\sqrt{|a|^2 + |b|^2}}$$

normalization

If the first qubit were measured to be "1", keep only those states and renormalize to get the new state:

$$|\psi'\rangle = \frac{c\,|10\rangle + d\,|11\rangle}{\sqrt{|c|^2 + |d|^2}}$$

Multi-qubit measurement

Measurement on a multi-qubit state:

1) Apply collapse principle to the measured state – keep only the amplitudes compatible with the measurement.
2) Renormalize the state probabilities, so the probabilities sum to 1.

For example, consider the state:

$$|\psi\rangle = \frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle + \frac{1}{\sqrt{2}}|3\rangle|8\rangle$$

Note: multiple qubits in each register $|x\rangle\,|y\rangle$

Imagine that we measured the first register. There are two possible outcomes: "1" and "3". What are the probabilities?

Multi-qubit measurement

$$|\psi\rangle = \frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle + \frac{1}{\sqrt{2}}|3\rangle|8\rangle$$

Probability of measuring "1":

$$P_1 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

Probability of measuring "3":

$$P_3 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

Multi-qubit measurement

$$|\psi\rangle = \frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle + \frac{1}{\sqrt{2}}|3\rangle|8\rangle$$

If the measurement outcome "1" were measured, the (unnormalized) state compatible with this is:

$$\frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle$$

After normalization, the state becomes:

$$|\psi'\rangle = \frac{\frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle}{\sqrt{\frac{1}{2}^2 + \frac{1}{2}^2}}$$

$$= \frac{|1\rangle|5\rangle + |1\rangle|6\rangle}{\sqrt{2}}$$

Multi-qubit measurement

$$|\psi\rangle = \frac{1}{2}|1\rangle|5\rangle + \frac{1}{2}|1\rangle|6\rangle + \frac{1}{\sqrt{2}}|3\rangle|8\rangle$$

If the measurement outcome “3” were measured, the (unnormalized) state compatible with this is:

$$\frac{1}{\sqrt{2}}|3\rangle|8\rangle$$

After normalization, the state becomes:

$$|\psi'\rangle = \frac{\frac{1}{\sqrt{2}}|3\rangle|8\rangle}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2}}$$

$$= |3\rangle|8\rangle$$

7.6 The Toffoli Gate

MULT20015

Lecture 7

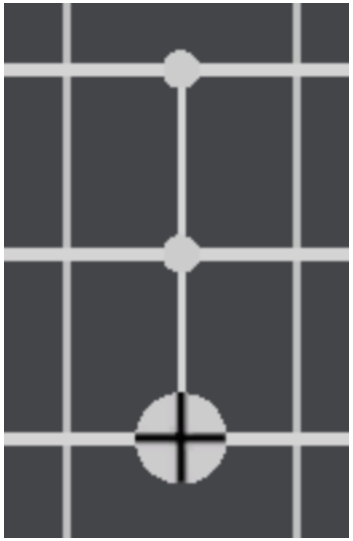
Toffoli gate

Toffoli gate plus NOT is universal for classical computation. It was used in the proof that classical computation can be made reversible!

Control qubit

Control qubit

Target qubit



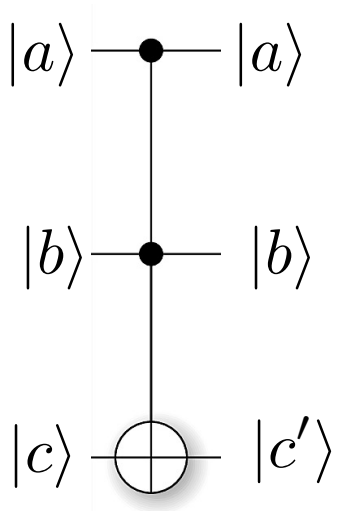
How states transform:

$$|000\rangle \to |000\rangle$$
$$|001\rangle \to |001\rangle$$
$$|010\rangle \to |010\rangle$$
$$|011\rangle \to |011\rangle$$
$$|100\rangle \to |100\rangle$$
$$|101\rangle \to |101\rangle$$
$$|110\rangle \to |111\rangle$$
$$|111\rangle \to |110\rangle$$

Rule: the target is flipped iff **both** the control qubits are in "1" state.

$$a\,|000\rangle + b\,|001\rangle + c\,|010\rangle + d\,|011\rangle$$
$$e\,|100\rangle + f\,|101\rangle + \boxed{g\,|110\rangle + h\,|111\rangle}$$
$$\to a\,|000\rangle + b\,|001\rangle + c\,|010\rangle + d\,|011\rangle$$
$$e\,|100\rangle + f\,|101\rangle + \boxed{h\,|110\rangle + g\,|111\rangle}$$

Toffoli Gate



Toffoli gate

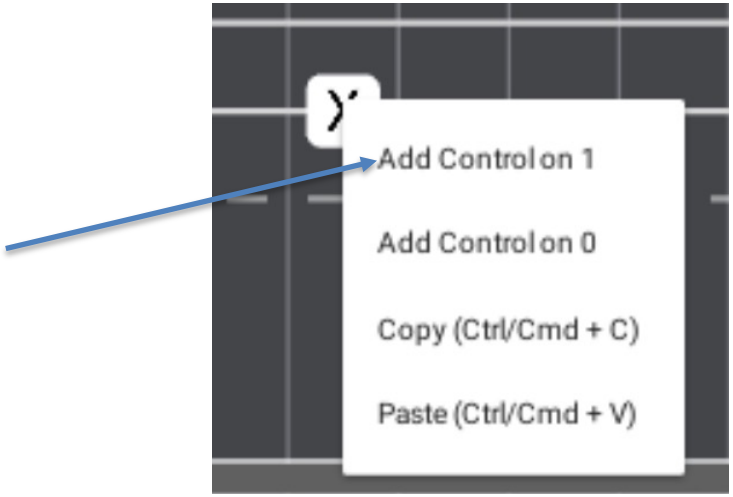
Control-Control-NOT gate

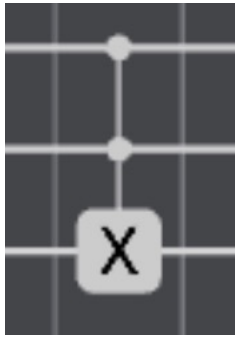
a	b	c	c’
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Multiply Controlled operations in QUI

You can add controls on single qubit gates in QUI.

Right click on a gate and select *"Add Control on 1"*





Click on the qubit where you would like a control. Repeat for as many controls as you would like.

Week 4

Lecture 7

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Lecture 8

8.1 Classical digital logic and universality
8.2 Reversible logic
8.3 Arithmetic operations on a quantum computer
8.4 Universality in quantum computing

Practice class 4

Multi-qubit states and operations

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5 – Classical complexity and simple quantum algorithms
6 – Cryptography and Shor's quantum factoring algorithm
7 – Post quantum cryptography and quantum key distribution
8 – Quantum search algorithms
9 – Quantum algorithms for option pricing
10 – Optimisation problems on quantum computers
11 – Portfolio optimisation using quantum computers
12 – Quantum machine learning